

ODD 2024
Tutorial Sheet - 10

Software Development Fundamentals – I (15B11CI111)

Course Outcomes (CO)	
CO1	Explain various phases of software development life cycle
CO2	Explain various data types, memory allocation schemes, precedence of arithmetical and logical operations, and need of array, and structures
CO3	Draw the flow chart and write the high level code for different problems
CO4	Apply and implement functions with or without pointers for different problems
CO5	Demonstrate and implement various operations like traverse, insertion, deletion, <i>etc.</i> on files

Note: Students are advised to submit their solutions to their respective tutorial faculty.

Q1. [CO4] Write a program to find the common elements between two arrays using a user-defined function. The arrays can have different sizes, and the result should not contain duplicate elements.

Q2. [CO4] Case Study: Matrix Operations in Scientific Computing

Problem:

You are part of a scientific research team working on simulating climate data. One of the tasks requires you to multiply two matrices representing temperature and pressure data over different regions of the Earth. Your team relies heavily on matrix operations to handle these data. Write a C program that accepts two matrices and performs matrix multiplication.

Requirements:

1. The program should read matrix dimensions and values.
2. Validate that the matrices can be multiplied (i.e., the number of columns of the first matrix equals the number of rows of the second matrix).
3. Display the resulting matrix after multiplication.

Q3.[CO4] Write a program to check if two given strings are anagrams of each other using a user-defined function. Ignore the case and spaces while checking.

Q4.[CO4 Case Study: Calculator for Financial Application

Problem:

You are designing a financial calculator that can perform basic arithmetic operations like addition, subtraction, multiplication, and division for financial data analysis. Based on the user's input, the appropriate operation should be applied to two numbers. The calculator make use of function pointers.

Requirements:

1. Input two numbers.

2. Perform the chosen arithmetic operation (addition, subtraction, multiplication, or division).
3. Handle division by zero gracefully.

Q5.[CO4] Case Study: Sorting and Ranking System

Problem:

You are tasked with developing a ranking system for a sports competition where participants' scores are stored in an array. Depending on the situation, the system should sort the scores in either ascending or descending order to display the rankings. Implement a program where you can select the sorting order using a function pointer.

Requirements:

1. Input a list of scores.
2. Choose between sorting in ascending or descending order.
3. Display the sorted list based on the user's choice.

Q6.[CO4] Case Study: Text Processing Application

Problem:

You are working on a text processing tool for a publishing company. The tool needs to modify text documents by converting the text either to uppercase or lowercase based on the user's preference. Implement a solution using function pointers.

Requirements:

1. Accept a string from the user.
2. Convert the string either to uppercase or lowercase based on user input.
3. Display the modified string.

Q7. [CO4] Write a program that dynamically allocates memory for an array using malloc(), calloc(), or realloc(). Use a function pointer to choose the appropriate memory allocation function at runtime based on the user's choice.

Q8. [CO4] Write a program that simulates a basic shape calculator where the user can calculate the area of a circle, square, or rectangle using function pointers. Define different functions for each shape and use a function pointer to call the appropriate one based on the user's choice.

Q9. [CO4] Write a program to perform different operations (like adding, subtracting, and multiplying elements) on a matrix using an array of function pointers. The program should allow the user to select the operation, and the appropriate function should be invoked using the function pointer.